Model	Task	Precision	Recall	Δ-*Recall*
LLaMA	I	74.5	52.5	*0*
	I^{GenArg}	70.9	57.3	*+4.8*
	$I^{RandArg\downarrow}$	66.9	**60.5**	*+8.0*
	$I^{RandArg\uparrow}$	70.6	51.5	*-1.0*
GPT-3.5	I	80.6	**96.5**	*0*
	I^{GenArg}	79.7	91.3	*-5.2*
	$I^{RandArg\downarrow}$	80.1	82.5	*-14.0*
	$I^{RandArg\uparrow}$	81.9	80.5	*-16.0*
PaLM	I	90.3	**84.0**	*0*
	I^{GenArg}	92.3	71.5	*-12.5*
	$I^{RandArg\downarrow}$	87.8	82.5	*-1.5*
	$I^{RandArg\uparrow}$	88.2	82.0	*-2.0*

Table 6: Scoring model outputs in different conditions of RTE-1. We indicate the **highest** and lowest recall score across replacement settings.

throughout, in order to better explore the abilities of these LLMs.

B RTE-1 Results For Experiment 2: Entities are Indices to Memory

The RTE-1 dataset contains complex natural language statements with varied linguistic features, so predictions about entailment are not decidable only on the basis of contained predicates. However, RTE-1 is a difficult challenge set for models, and interesting to compare to in the broader domain of NLI. Though the sentences are much more complex, we are able to conduct an analogous experiment as in §6 by first identifying spans of named entities and their respective entity types, then replacing the entities with new ones. As before, we compare model scores on the original dataset to three test conditions: generic arguments ("location X", "person Y", etc.), sampled low-frequency entities constrained to the same type, and the same for high-frequency sampled entities. Since only the entities in each statement have been altered, the entailment labels between premise/hypothesis pairs remain unchanged, and an ideal model capable of generalizing inference would make the same predictions across dataset conditions. Results are shown in Table 6.

We observe similar trends to those reported on Levy/Holt. GPT-3.5 performs very consistently between Levy/Holt and RTE-1 in terms of degrading recall when information is changed in each sample. We observe that model performance is worse than the original dataset when using generic arguments, and worse still using type-constrained random arguments. We further observe that across all three LLMs across both datasets, models consistently achieve worse recall using high-frequency entities than low-frequency entities, supporting the claim that increasing the frequency of entity occurrence in training data impedes generalization.

Different from in Levy/Holt, we observe some noise in LLaMA's predictions; the recall on the original task is actually lower than the generic argument condition and the low-frequency entity condition. We note that overall, LLaMA is the weakest LLM tested in this experiment on both Levy/Holt and RTE-1, and that its performance on RTE-1 is particularly low. We suggest that the increased difficulty of RTE-1 over Levy/Holt (due to having much more linguistic variation) is simply too complex for LLaMA, which is neither the largest LLM tested, nor instruction-finetuned.

We also observe a smaller gap between PaLM's recall rates across dataset conditions, though the gaps are consistent with our claims. While the model appears able to generalize to conditions in which random real arguments are inserted, recall on the generic argument condition is significantly degraded. Failure on this control condition indicates that the model may not be generalizing as well as the other conditions would imply.

C The Ineffectiveness of Instructing LLMs to Stop Conditioning on Attested Information

In §5 and §6, we showed that entailment predictions from LLMs are strongly biased by their predictions on the attestation of hypotheses. We wondered whether there are intuitive prompt engineering techniques to steer its behavior away from attending to attestation.

Towards this goal, we experimented with prepending a brief task description to the few-shot prompts in part B of Table 13, explicitly instructing the models to ignore the attestedness of individual statements: *Please check the entailments between the following hypothetical statements. Ignore the veracity of these statements.*

We replicated the experiments in §5 and §6 with GPT-3.5, since GPT-3.5 is an instruction-finetuned model trained to be responsive to prompts, where the other two LLM families are only pre-trained. Despite having been instruction-finetuned, the results with GPT-3.5 show only marginal improvements in model behavior.

task	GPT-3.5	Instructed to Ignore Attestedness	Not Instructed
I	$P(\texttt{Entail} \mid \texttt{Attested})$	74.3	77.6
I	$P(\texttt{Entail} \mid \neg\texttt{Attested})$	57.8	63.6
$I_{RandPrem}$	$P(\texttt{Entail} \mid \texttt{Attested})$	39.0	41.3
$I_{RandPrem}$	$P(\texttt{Entail} \mid \neg\texttt{Attested})$	17.6	18.8

Table 7: We estimate the probability of positive predictions of `Entail` in I and $I_{RandPrem}$ tasks respectively given that the hypothesis is attested, namely $\Lambda = \texttt{Attested}$. **Not instructed** results are copied from Figure 2 and listed here for ease of comparison; also note that all $I_{RandPrem} = \texttt{Entail}$ predictions are false positives.

GPT-3.5 Condition	Task	Levy/Holt (Directional)		
		Precision	Recall	**Δ-Recall**
Few-shot, instructed to ignore attestedness.	I	64.9	**90.8**	0
	I^{GenArg}	73.5	69.3	-21.5
	$I^{RandArg\downarrow}$	64.6	68.4	-22.4
	$I^{RandArg\uparrow}$	67.5	58.1	-32.7
Few-shot, no instructions.	I	62.4	**92.3**	0
	I^{GenArg}	65.1	75.7	-16.6
	$I^{RandArg\downarrow}$	65.5	66.5	-25.8
	$I^{RandArg\uparrow}$	68.8	55.3	-37.0

Table 8: GPT-3.5 predictions when models are explicitly instructed to avoid taking the attestedness of individual statements into account. In the upper half are the instructed behavior, and in the lower half are the regular few-shot behavior as in Table 4. Differences in recalls remain at a similar scale, with precision again stable, where the benefit from the explicit instruction is marginal.

In Table 7, we show that instructing GPT-3.5 to ignore attestation does not help narrow the gap between $\Lambda = \texttt{Attested}$ and $\Lambda = \neg\texttt{Attested}$; instead, probabilities of predicting `Entail` went down by similar amounts, indicating that the model is becoming slightly more conservative in predicting positives when instructed to ignore attestation, but not in a principled manner.

Further, as shown in Table 8, despite the explicit instruction, recall still drops at similar scales when arguments are randomly replaced with the same sets of frequent/infrequent replacement entities as before. Since GPT-3.5 has been instruction-finetuned to respond to prompts, its failure means eradicating such biases from model outputs is a difficult task, one that needs further research attention.

D Statistics of Consistency Subsets

The statistics of consistency subsets are presented in Table 9.

E The Reliability of Λ Measure and Its Relation to Consensus of Attestation

The Λ-consistency subsets most directly capture the impacts of the *attestation bias*. However, these subset separations are based on Λ predictions from individual models, which can be noisy, subject to model-specific idiosyncracies such as trigger words or certain syntactic structures in the prompt, etc.

To verify that the performance gaps in Λ-consistency subsets that we observe in §8 comes from predicted attestedness and not some idiosyncrasy, we experiment with another pair of subsets based on *consensus attestation* instead of individually *predicted attestation*.

We use a majority vote among the three independently-trained LLMs to approximate *consensus attestation*. The approximation is denoted as $\tilde{\Lambda}$. This is because any model-specific idiosyncrasies should not be shared between LLMs independently trained from different source corpora in general. Therefore, with the majority vote, we reduce this noise and acquire predictions on the *consensus attestation* of statements.

Performances of LLMs between $\tilde{\Lambda}$-consistency subsets are listed in Table 10. Gaps between the $\tilde{\Lambda}$-consistency subsets that are larger than Λ-consistency gaps are colored red; those narrower than Λ-consistency gaps are colored green. It is clear that the gaps are consistent between Λ/$\tilde{\Lambda}$-consistency experiments, where the gaps are even larger on many occasions. This confirms that the

# of Entries	Levy/Holt			RTE-1		
	LLaMA	GPT-3.5	PaLM	LLaMA	GPT-3.5	PaLM
$V_{\text{CONSISTENT}}$	955	947	999	479	447	480
$V_{\text{ADVERSARIAL}}$	829	837	785	321	353	320
$F_{\text{CONSISTENT}}$	972			286		
$F_{\text{ADVERSARIAL}}$	220			247		

Table 9: Subsets defined by the consistency between entailment label L and either Λ (hypothesis attestation prediction from each LLM) or Φ (model-agnostic relative frequency bias). CONSISTENT subsets are where L agrees with Λ/Φ. ADVERSARIAL subsets are where L disagrees with Λ/Φ.

Model	Task	Levy/Holt		
		$\tilde{\Lambda}_{cons.}$	$\tilde{\Lambda}_{adv.}$	diff.
LLaMA	I	65.3	6.5	***-58.8***
GPT-3.5	I	70.8	23.5	***-47.3***
PaLM	I	80.7	28.3	***-52.4***
LLaMA	I^{GenArg}	54.4	29.6	***-24.8***
GPT-3.5	I^{GenArg}	56.2	35.5	***-20.7***
PaLM	I^{GenArg}	59.3	40.1	***-19.2***

Table 10: LLM performance on Levy/Holt subsets where Attestation $\tilde{\Lambda}$ is Consistent/Adversarial to the labels, measured with AUC_{norm} (0% = random chance performance). Performance drops from $\tilde{\Lambda}_{cons}$ to $\tilde{\Lambda}_{adv}$ are presented in the *diff.* columns, sharper decreases than Λ-comparisons in Table 5 are colored **red**, milder ones are colored **green**.

performance gaps in Λ-consistency experiments can be credited to the *attestation bias*, rather than model-specific idiosyncrasies.

It is also to be noted that, since the Φ-consistency subsets are separated based on the model-agnostic criterion Φ, model-specific idiosyncrasies are not a problem for Φ-consistency comparisons.

F Impacts of Bias on GPT-4 Performance

GPT-4 (OpenAI, 2023) is a recent, strong LLM claiming SOTA performance on various NLP tasks. Due to its closed-source nature and the impossibility of fully tracking the sources of its behaviors, we refrain from reporting results with it in the main content of this paper.

However, in order to provide a richer context for the *attestation bias* and the *relative frequency bias*, in this section we report the performance differences of GPT-4 between subsets consistent/adversarial to the two biases.

As a light-weight experiment, we elicit GPT-4 predictions in the original I task in the zero-shot setting, and re-use subsets from experiments in §8. Specifically, for the *attestation bias*, we use the majority vote $\tilde{\Lambda}$ among LLaMA, GPT-3.5 and PaLM, to approximate Λ predictions from GPT-4 itself; for the *relative frequency bias*, we keep the Φ measure for approximating corpus-frequency of terms.

Because GPT-4 is a commercial service and does not provide logit confidence with their discrete predictions, AUC_{norm} values could not be calculated. Therefore, we are forced to report **the *F-1 scores* at the binary prediction point of confidence**. As results in Table 12 show, we observe the same trend as in §8: for the subset adversarial to each factor, GPT-4 performance also drops substantially.

This experiment is designed to provide more context for the two biases discussed in the paper and **NOT** to compare GPT-4 with other models; however, we can conclude that GPT-4 is subject to the same fragilities as the other LLMs w.r.t. the two biases, where our conclusions and recommendations also apply.

G Dataset Statistics and Dev Set Performance

In the paper, we have examined the behavior and performance of three major LLM families on two NLI datasets: Levy/Holt and RTE-1.

The directional portion of Levy/Holt dataset[9] contains 630 entries in its dev set, and 1784 entries in its test set; the RTE-1 dataset[10] contains 567 entries in its dev set, and 800 entries in its test set. Each dataset has a 50%/50% class distribution between `Entail` and `No-Entail` (for RTE-1 dev set, the numbers of entries in the two label classes differ by 1).

In Table 11, we report dev set performance and the best prompt template used for each model on each dataset. Note that no training is involved in

[9] https://github.com/mjhosseini/entgraph_eval/tree/master/LevyHoltDS

[10] https://www.kaggle.com/datasets/nltkdata/rte-corpus?resource=download